Scandium

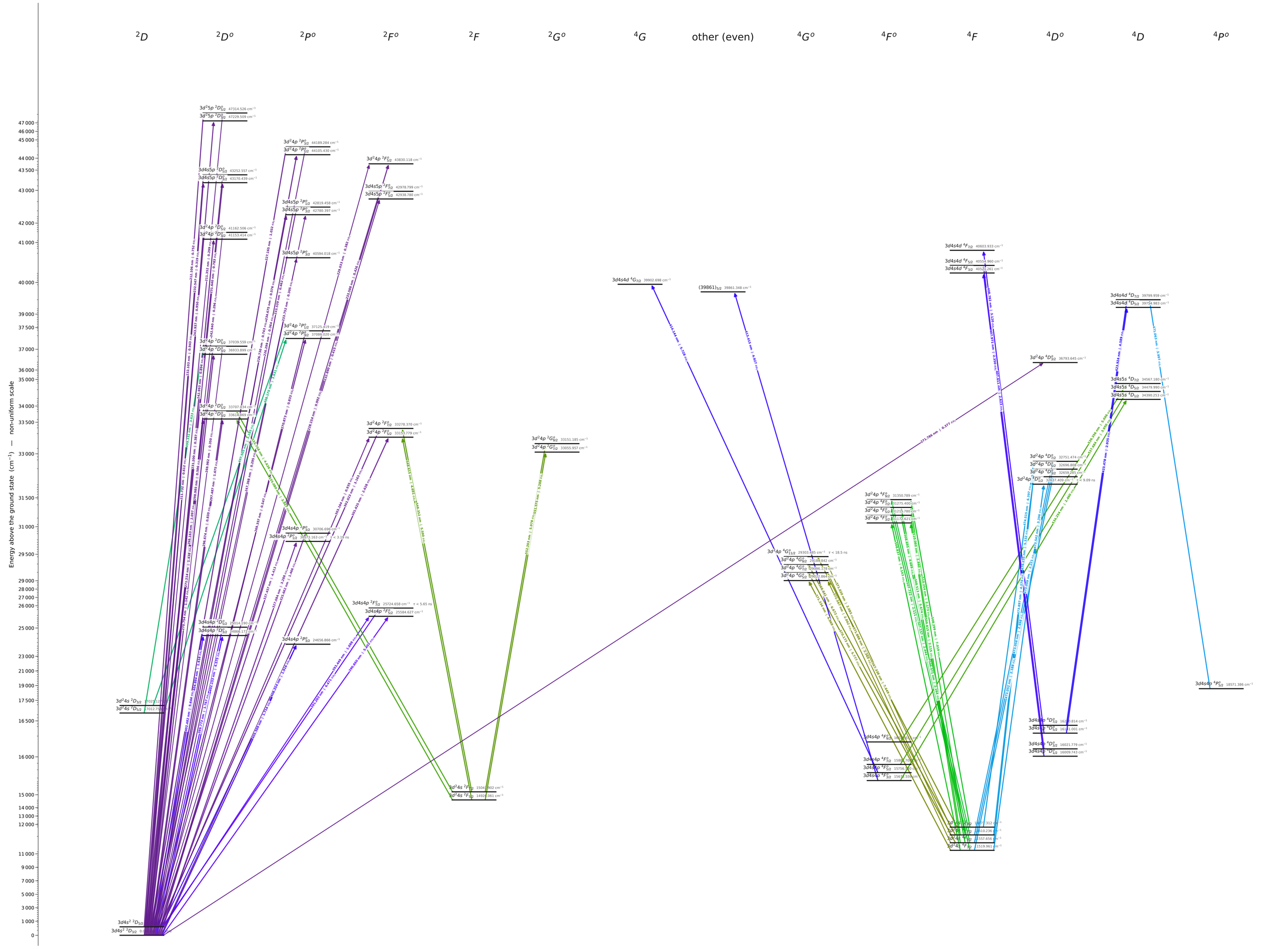
Sc (Sc I). Strongest classified transitions below 2 μm (NIST): 74 levels, 90 lines.

Sc$^+$ ionisation limit 52922.00 cm^{-1} = 6.56149 eV (188.957 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$3d^24s\ ^4F_{9/2}$ – $3d^24p\ ^4G^o_{11/2}$	567.3381	567.1807	528.4194	7.649	5.410e+07	8.612	100	laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{45}Sc hyperfine structure (I = 7/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3d4s^2\ ^2D_{3/2}$	269.5581(9)	26.361(9)	F=2: -1805.39 F=3: -1015.55 F=4: +55.15 F=5: +1421.77
$3d4s^2\ ^2D_{5/2}$	-37.372(15)	-1.246.64	F=1: -1246.64 F=2: -1017.90 F=3: -678.79 F=4: -234.11 F=5: +309.72 F=6: +944.70
$3d^24s\ ^4F_{3/2}$	-158.512(3)	-5.189(21)	F=2: +1067.18 F=3: +595.35 F=4: -37.22 F=5: -833.49
$3d^24s\ ^4F_{5/2}$	154.0352(21)	-6.45(3)	F=1: -1736.35 F=2: -1426.44 F=3: -962.26 F=4: -344.64 F=5: +425.30 F=6: +1346.20
$3d^24s\ ^4F_{7/2}$	250.0119(21)	-9.08(3)	F=0: -3942.55 F=1: -3691.61 F=2: -3189.92 F=3: -2437.85 F=4: -1435.95 F=5: -184.96 F=6: +1314.18 F=7: +3060.38
$3d^24s\ ^4F_{9/2}$	285.966(9)	-15.5(7)	F=1: -5511.94 F=2: -4937.79 F=3: -4077.04 F=4: -2930.23 F=5: -1498.10 F=6: +218.43 F=7: +2218.26 F=8: +4500.10
$3d^24s\ ^2F_{5/2}$	292(6)	–	F=1: -3281.60 F=2: -2698.21 F=3: -1823.11 F=4: -656.32 F=5: +802.17 F=6: +2552.36
$3d^24s\ ^2F_{7/2}$	-18.0(12)	–	F=0: +283.30 F=1: +265.32 F=2: +229.34 F=3: +175.38 F=4: +103.43 F=5: +13.49 F=6: -94.44 F=7: -220.35
$3d4s4p\ ^4F^o_{3/2}$	-137.6(3)	-27(4)	F=2: +914.54 F=3: +520.78 F=4: -22.01 F=5: -729.10
$3d4s4p\ ^4F^o_{5/2}$	289.6(3)	–	F=1: -3257.99 F=2: -2678.80 F=3: -1810.00 F=4: -651.60 F=5: +796.40 F=6: +2534.00
$3d4s4p\ ^4F^o_{7/2}$	359.5(3)	-39(5)	F=0: -5682.23 F=1: -5318.81 F=2: -4592.75 F=3: -3505.64 F=4: -2059.89 F=5: -258.65 F=6: +1894.08 F=7: +4393.53
$3d4s4p\ ^4D^o_{1/2}$	-510.8(18)	–	F=3: +1149.40 F=4: -893.98
$3d4s4p\ ^4D^o_{3/2}$	342.4(24)	–	F=2: -2310.95 F=3: -1283.86 F=4: +85.59 F=5: +1797.41
$3d4s4p\ ^4F^o_{9/2}$	390.9(3)	-63(14)	F=1: -7554.38 F=2: -6763.49 F=3: -5579.03 F=4: -4003.26 F=5: -2039.20 F=6: +309.39 F=7: +3037.99 F=8: +6141.32
$3d4s4p\ ^4D^o_{5/2}$	306.1(6)	–	F=1: -3443.49 F=2: -2831.32 F=3: -1913.05 F=4: -688.70 F=5: +841.74 F=6: +2678.27
$3d4s4p\ ^4D^o_{7/2}$	457.8(3)	22(12)	F=0: -7198.36 F=1: -6742.81 F=2: -5831.26 F=3: -4462.83 F=4: -2636.16 F=5: -349.48 F=6: +2399.45 F=7: +5613.31
$3d^24s\ ^2D_{5/2}$	552.5(6)	89(11)	F=1: -6168.12 F=2: -5088.53 F=3: -3459.59 F=4: -1269.88 F=5: +1495.89 F=6: +4856.79
$3d^24s\ ^2D_{3/2}$	-288.4(9)	-17(8)	F=2: +1937.71 F=3: +1084.50 F=4: -64.31 F=5: -1518.30
$3d4s4p\ ^4P^o_{5/2}$	587.9(6)	–	F=1: -6613.80 F=2: -5438.01 F=3: -3674.33 F=4: -1322.76 F=5: +1616.71 F=6: +5144.06
$3d4s4p\ ^2P^o_{1/2}$	442(5)	–	F=3: -994.26 F=4: +773.32
$3d4s4p\ ^2P^o_{3/2}$	12.6(9)	-191(13)	F=2: -187.13 F=3: -13.17 F=4: +91.67 F=5: +18.44
$3d4s4p\ ^2D^o_{3/2}$	278(5)	–	F=2: -1875.88 F=3: -1042.15 F=4: +69.48 F=5: +1459.02
$3d4s4p\ ^2D^o_{5/2}$	31(4)	–	F=1: -347.38 F=2: -285.63 F=3: -192.99 F=4: -69.48 F=5: +84.92 F=6: +270.19
$3d4s4p\ ^2F^o_{5/2}$	171.5(15)	–	F=1: -1929.16 F=2: -1586.20 F=3: -1071.76 F=4: -385.83 F=5: +471.57 F=6: +1500.46
$3d4s4p\ ^2F^o_{7/2}$	125(3)	–	F=0: -1964.24 F=1: -1839.53 F=2: -1590.10 F=3: -1215.96 F=4: -717.10 F=5: -93.53 F=6: +654.75 F=7: +1527.74
$3d^24p\ ^4G^o_{5/2}$	279.1(12)	-29(7)	F=1: -3155.69 F=2: -2589.08 F=3: -1742.32 F=4: -619.18 F=5: +775.31 F=6: +2434.84
$3d^24p\ ^4G^o_{7/2}$	147.2(3)	–	F=0: -2318.37 F=1: -2171.17 F=2: -1876.78 F=3: -1435.18 F=4: -846.39 F=5: -110.40 F=6: +772.79 F=7: +1803.18
$3d^24p\ ^4G^o_{9/2}$	83.3(6)	-25(4)	F=1: -1615.74 F=2: -1445.51 F=3: -1190.89 F=4: -852.78 F=5: -432.36 F=6: +68.88 F=7: +649.16 F=8: +1306.42
$3d^24p\ ^4G^o_{11/2}$	44.7(3)	-39(7)	F=2: -1032.47 F=3: -892.03 F=4: -706.22 F=5: -476.24 F=6: -203.63 F=7: +109.76 F=8: +461.81 F=9: +850.06
$3d4s4p\ ^2P^o_{1/2}$	240(19)	–	F=3: -539.63 F=4: +419.71
$3d4s4p\ ^2P^o_{3/2}$	327(4)	–	F=2: -2209.77 F=3: -1227.65 F=4: +81.84 F=5: +1718.71
$3d^24p\ ^4F^o_{3/2}$	280.3(3)	–	F=2: -1892.07 F=3: -1051.15 F=4: +70.08 F=5: +1471.61
$3d^24p\ ^4F^o_{5/2}$	121.1(3)	–	F=1: -1362.56 F=2: -1120.32 F=3: -756.98 F=4: -272.51 F=5: +333.07 F=6: +1059.77
$3d^24p\ ^4F^o_{7/2}$	67.5(3)	–	F=0: -1062.39 F=1: -994.94 F=2: -860.03 F=3: -657.67 F=4: -387.86 F=5: -50.59 F=6: +354.13 F=7: +826.30
$3d^24p\ ^4F^o_{9/2}$	44.4(3)	–	F=1: -854.11 F=2: -765.37 F=3: -632.26 F=4: -454.79 F=5: -232.94 F=6: +33.28 F=7: +343.86 F=8: +698.82
$3d^24p\ ^4D^o_{1/2}$	432.9(9)	–	F=3: -974.03 F=4: +757.58
$3d^24p\ ^4D^o_{3/2}$	126.2(15)	–	F=2: -851.93 F=3: -473.30 F=4: +31.55 F=5: +662.62
$3d^24p\ ^4D^o_{5/2}$	70.5(9)	–	F=1: -792.58 F=2: -651.67 F=3: -440.32 F=4: -158.51 F=5: +193.74 F=6: +616.45
$3d^24p\ ^4D^o_{7/2}$	42.0(3)	–	F=0: -661.04 F=1: -619.07 F=2: -535.13 F=3: -409.22 F=4: -241.33 F=5: -31.48 F=6: +220.35 F=7: +514.14
$3d^24p\ ^2G^o_{9/2}$	169.4(12)	–	F=0: -2667.78 F=1: -2498.39 F=2: -2159.63 F=3: -1651.48 F=4: -973.95 F=5: -127.04 F=6: +889.26 F=7: +2074.94
$3d^24p\ ^2G^o_{7/2}$	71(4)	–	F=1: -1361.96 F=2: -1220.45 F=3: -1008.20 F=4: -725.20 F=5: -371.44 F=6: +53.06 F=7: +548.32 F=8: +1114.33
$3d^24p\ ^2F^o_{5/2}$	226.9(18)	–	F=1: -2553.11 F=2: -2099.22 F=3: -1418.39 F=4: -510.62 F=5: +624.09 F=6: +1985.75
$3d^24p\ ^2F^o_{7/2}$	39.9(6)	–	F=0: -627.99 F=1: -588.12 F=2: -508.37 F=3: -388.76 F=4: -229.27 F=5: -29.90 F=6: +209.33 F=7: +488.44
$3d^24p\ ^2D^o_{5/2}$	236(3)	–	F=2: -1590.55 F=3: -883.64 F=4: +58.91 F=5: +1237.09
$3d^24p\ ^2D^o_{3/2}$	41.7(15)	–	F=1: -468.80 F=2: -385.46 F=3: -260.44 F=4: -93.76 F=5: +114.60 F=6: +364.62
$3d4s5s\ ^4D_{1/2}$	-739.6(24)	–	F=3: +1664.07 F=4: -1294.28
$3d4s5s\ ^4D_{5/2}$	595.1(6)	–	F=1: -6694.74 F=2: -5504.56 F=3: -3719.30 F=4: -1338.95 F=5: +1636.49 F=6: +5207.02
$3d4s5s\ ^4D_{7/2}$	603.8(12)	–	F=0: -9509.57 F=1: -8905.78 F=2: -7698.22 F=3: -5886.88 F=4: -3471.75 F=5: -452.84 F=6: +3169.86 F=7: +7396.33
$3d^24p\ ^4D^o_{1/2}$	56(4)	–	F=2: -376.39 F=3: -209.10 F=4: +13.94 F=5: +292.75
$3d^24p\ ^2D^o_{3/2}$	162.2(15)	–	F=2: -1094.77 F=3: -608.20 F=4: +40.55 F=5: +851.49
$3d^24p\ ^2D^o_{5/2}$	133.1(9)	–	F=1: -1497.46 F=2: -1231.25 F=3: -831.92 F=4: -299.49 F=5: +366.05 F=6: +1164.69
$3d^24p\ ^2P^o_{3/2}$	215.6(9)	–	F=2: -1454.97 F=3: -808.32 F=4: +53.89 F=5: +1131.64
$3d^24p\ ^2P^o_{1/2}$	110.3(15)	–	F=3: -248.23 F=4: +193.07
$3d4s4d\ ^4D_{5/2}$	440.4(18)	–	F=1: -4954.44 F=2: -4073.66 F=3: -2752.47 F=4: -990.89 F=5: +1211.09 F=6: +3853.46
$3d4s4d\ ^4D_{7/2}$	503.1(6)	–	F=0: -7923.06 F=1: -7420.01 F=2: -6413.91 F=3: -4904.75 F=4: -2892.55 F=5: -377.29 F=6: +2641.02 F=7: +6162.38
$(39861)_{5/2}$	-291.1(18)	–	F=1: +3274.86 F=2: +2692.66 F=3: +1819.37 F=4: +654.97 F=5: -800.52 F=6: -2547.11
$3d4s4d\ ^4G_{7/2}$	101.3(21)	–	F=0: -1595.94 F=1: -1494.62 F=2: -1291.96 F=3: -987.97 F=4: -582.65 F=5: -76.00 F=6: +531.98 F=7: +1241.29
$3d4s4d\ ^4F_{3/2}$	-215(10)	–	F=2: +1452.94 F=3: +807.19 F=4: -53.81 F=5: -1130.07
$3d4s4d\ ^4F_{5/2}$	188(4)	–	F=1: -2118.03 F=2: -1741.49 F=3: -1176.68 F=4: -423.61 F=5: +517.74 F=6: +1647.36
$3d4s5p\ ^2P^o_{3/2}$	136(4)	–	F=2: -918.71 F=3: -510.40 F=4: +34.03 F=5: +714.55
$3d4s4d\ ^4F_{7/2}$	279.4(15)	–	F=0: -4400.65 F=1: -4121.25 F=2: -3562.43 F=3: -2724.21 F=4: -1606.59 F=5: -209.56 F=6: +1466.88 F=7: +3422.73
$3d^24p\ ^2D^o_{3/2}$	-195(4)	–	F=2: +1317.36 F=3: +731.87 F=4: -48.79 F=5: -1024.62
$3d^24p\ ^2D^o_{5/2}$	469.2(18)	–	F=1: -5278.22 F=2: -4339.87 F=3: -2932.34 F=4: -1055.64 F=5: +1290.23 F=6: +4105.28
$3d4s5p\ ^2P^o_{3/2}$	758(9)	–	F=2: -5117.68 F=3: -2843.16 F=4: +189.54 F=5: +3980.42
$3d4s5p\ ^2F^o_{7/2}$	67(27)	–	F=0: -1057.67 F=1: -990.51 F=2: -856.21 F=3: -654.75 F=4: -386.13 F=5: -50.37 F=6: +352.56 F=7: +822.63
$3d4s5p\ ^2D^o_{3/2}$	342(4)	–	F=2: -2310.95 F=3: -1283.86 F=4: +85.59 F=5: +1797.41
$3d4s5p\ ^2D^o_{5/2}$	219(11)	–	F=1: -2468.79 F=2: -2029.89 F=3: -1371.55 F=4: -493.76 F=5: +603.48 F=6: +1920.17
$3d^24p\ ^2F^o_{5/2}$	11(5)	–	F=1: -128.16 F=2: -105.38 F=3: -71.20 F=4: -25.63 F=5: +31.33 F=6: +99.68
$3d^24p\ ^2P^o_{1/2}$	463(28)	–	F=3: -1040.80 F=4: +809.51

Measured A, B constants; sources in the data file.